hw06

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

install.packages("bio3d")

```
library(bio3d)
```

Can you improve this analysis code?

First, I will load the bio3d package and run the repetitive code to understand its output.

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```

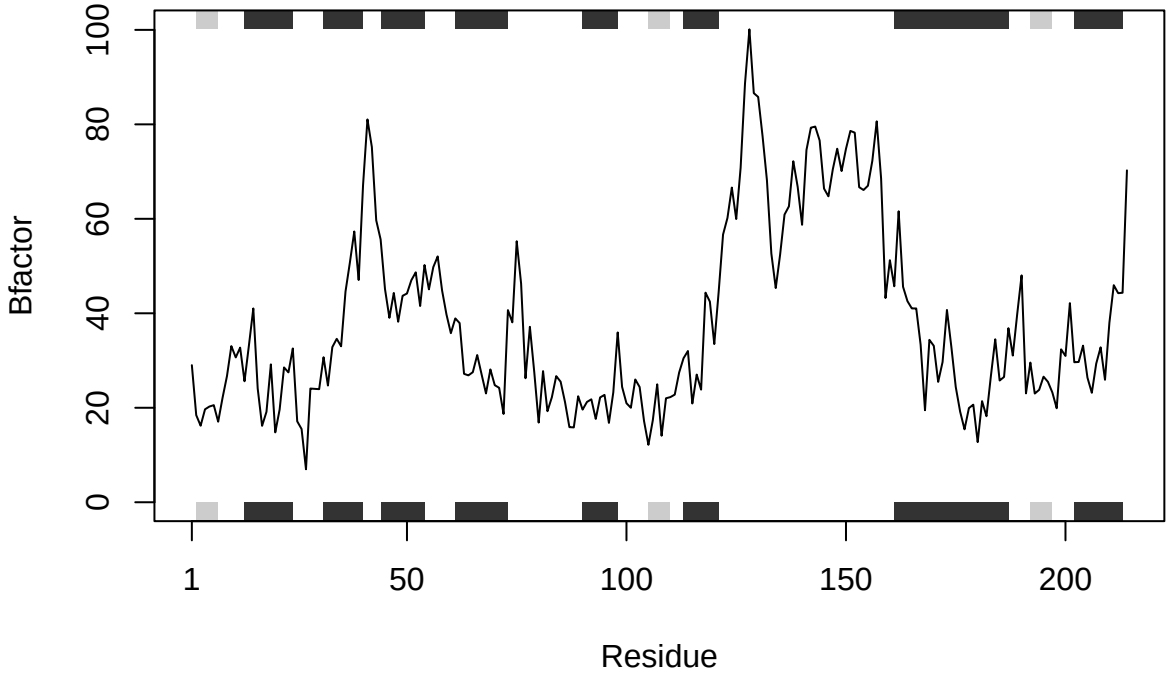
```
s2 <- read.pdb("1AKE") # kinase no drug
```

```
##   Note: Accessing on-line PDB file
##    PDB has ALT records, taking A only, rm.alt=TRUE
```

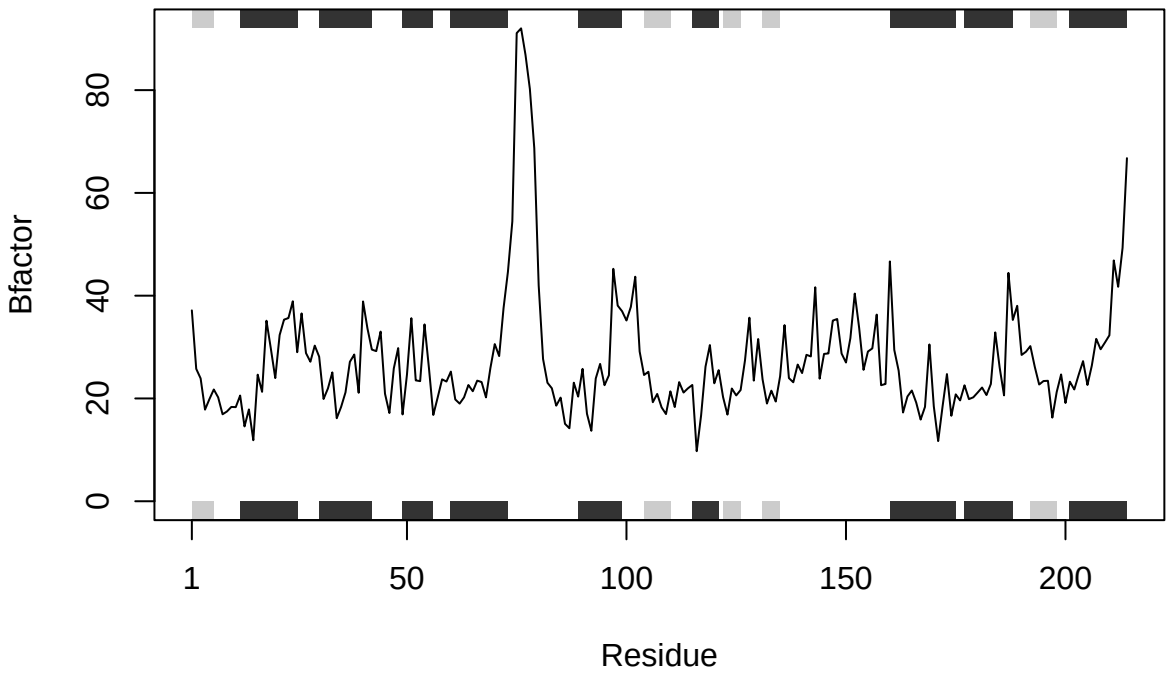
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```

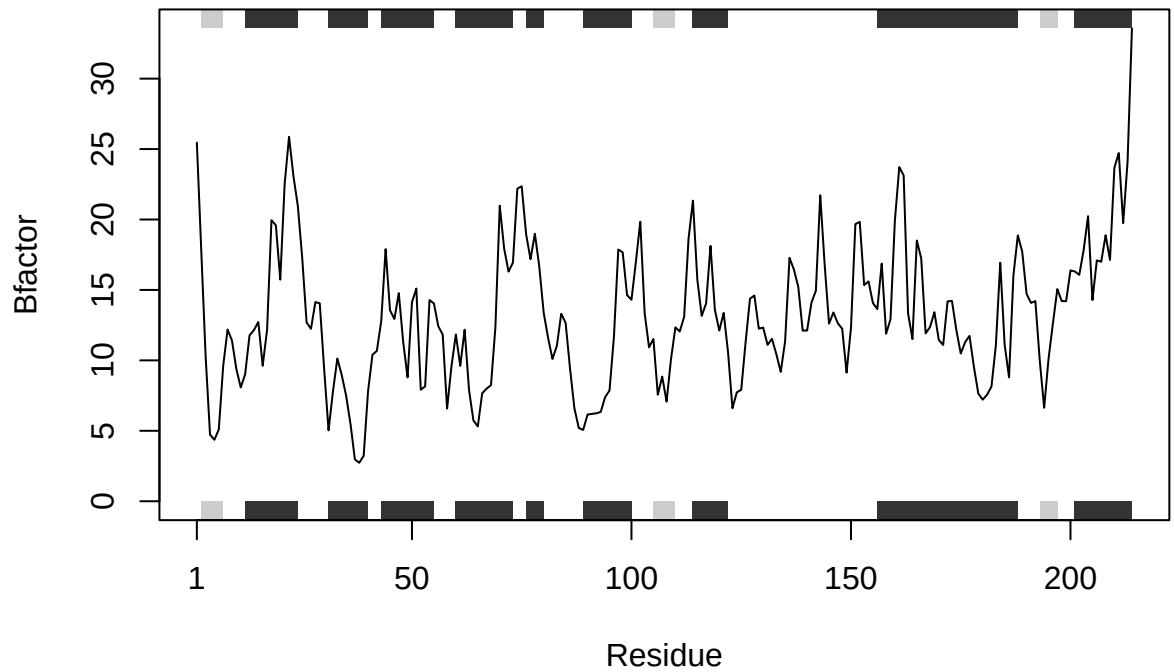
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s3, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```




```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```




```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



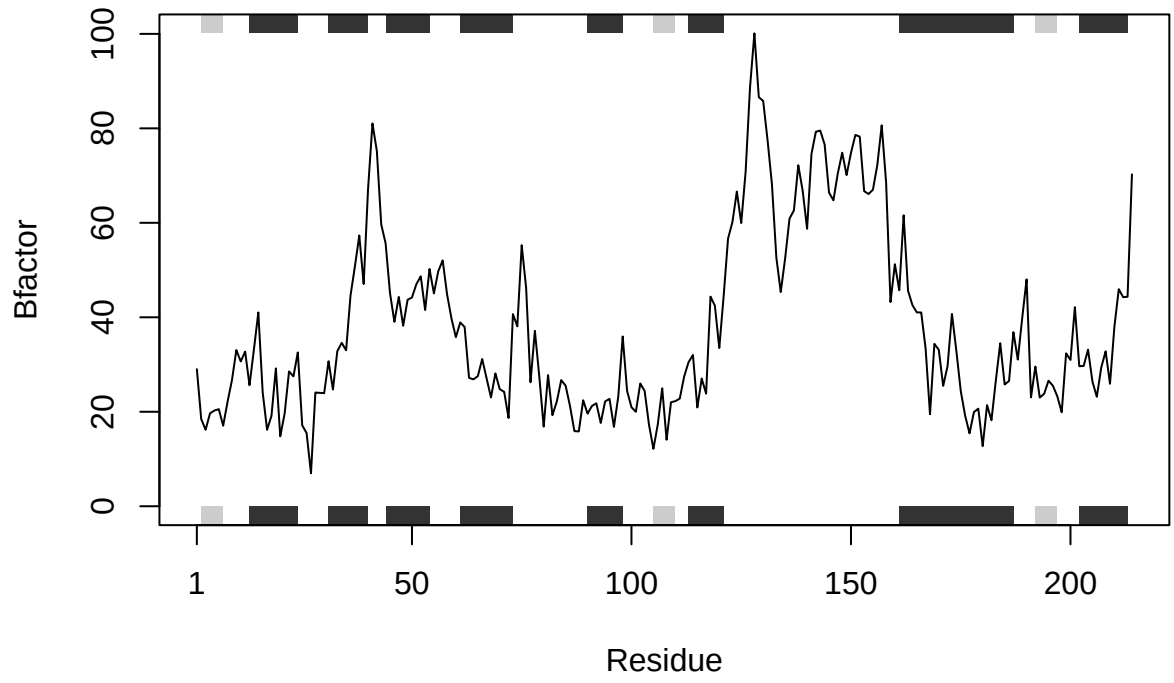
Since the the code written above is highly repetitive, I will create a function that cuts out all of the repetition used to make these plots. The input for my function will be the `pdb file name` for example, "4AKE". Once the input arguement is read, the function will create a dataframe of that file using `read.pdb`, editing the file using `trim.pdb`, create a variable for `atom$b` in that file, and finally creating a plot using `plotb3` by accessing all of the variables previously created. When running the function, the same plots are spit out as the highly repatitive code above.

```
Bioplot <- function(x){
  pdb <- read.pdb(x)
  pdb.chaina <- trim.pdb(pdb, chain="A", elety="CA")
  pdb.atomb <- pdb.chaina$atom$b
  plotb3(pdb.atomb, sse=pdb.chaina, typ="l", ylab="Bfactor")
}
```

```
s1 <- Bioplot("4AKE")
```

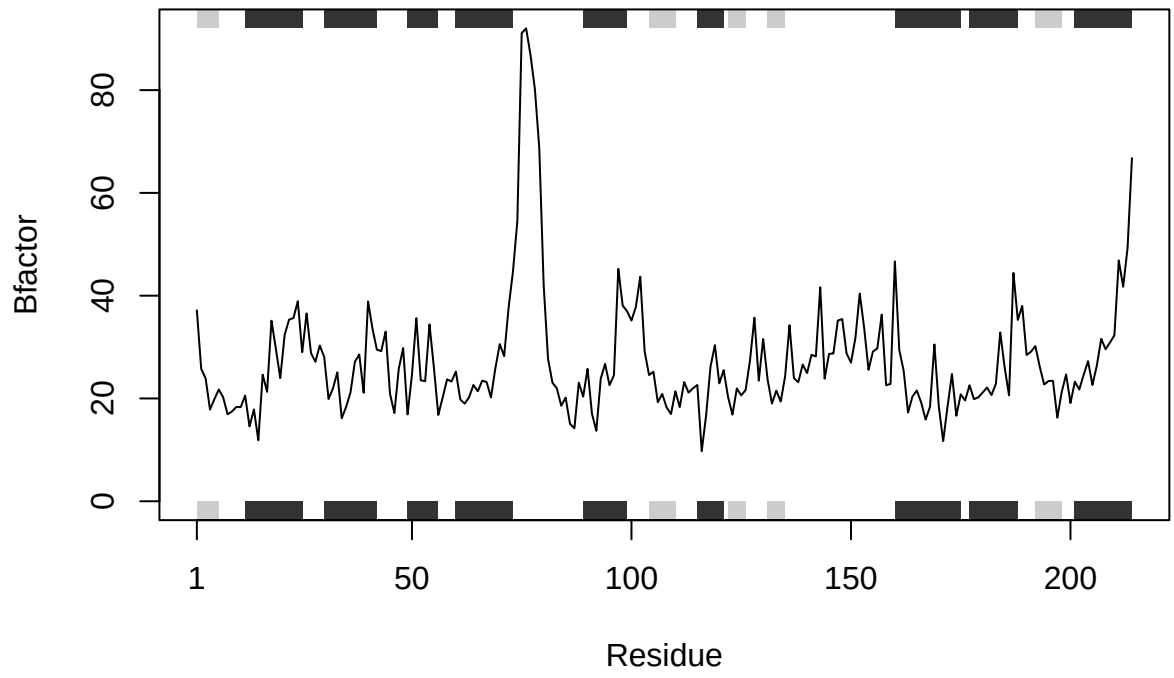
```
##   Note: Accessing on-line PDB file
```

```
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## /var/folders/sg/5mg47d_13s535w6ym_bg45fc0000gn/T//RtmpuYkSBq/4AKE.pdb exists.
## Skipping download
```



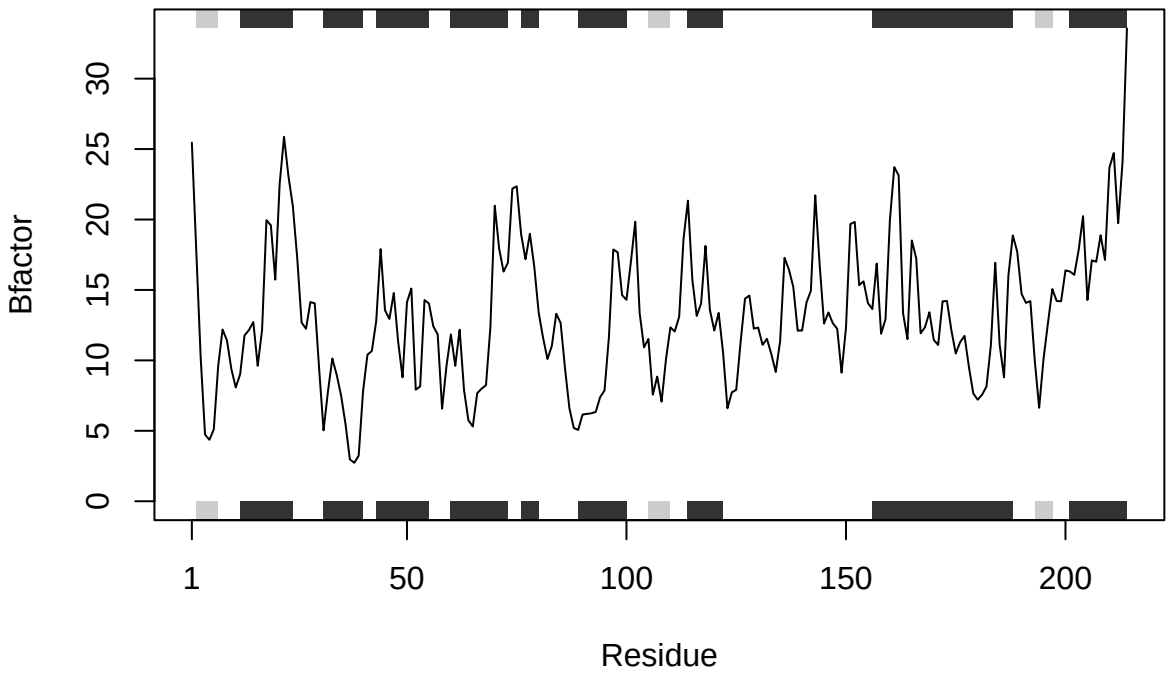

```r
s2 <- Bioplot("1AKE")
```

```
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## /var/folders/sg/5mg47d_13s535w6ym_bg45fc0000gn/T//RtmpuYkSBq/1AKE.pdb exists.
## Skipping download

##    PDB has ALT records, taking A only, rm.alt=TRUE
```




```r
s3 <- Bioplot("1E4Y")
```

```
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## /var/folders/sg/5mg47d_13s535w6ym_bg45fc0000gn/T//RtmpuYkSBq/1E4Y.pdb exists.
## Skipping download
```


Bfactor
0
5
10
15
20
25
30
1
50
100
150
200
Residue